

Interrelation of Gate Resistance and Emitter/Source Inductance Impact on Inductive Load Phase-Leg Crosstalk

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Abstract—Crosstalk poses a significant concern in power electronics converters that incorporate a phase-leg structure. The detrimental impact of crosstalk can result in device malfunction or failure. In this article, a model of the phase-leg structure emphasizing parasitic elements is introduced. Subsequently, the mathematical model of the circuit is deduced, followed by an investigation into the impact of low-side insulated gate bipolar transistor (IGBT) gate resistance and the sensitivity of low-side switch emitter inductance. We explored in this article that in the inductive load switching a negative spike also happens in the lower device gate terminal during the crosstalk prior to the positive spike. This negative spike can be harmful for the gate oxide insulator of the device. For optimal outcomes, a double-pulse test is set up. This test entails the application of two closely spaced voltage pulses to a device, allowing the assessment of its switching characteristics in inductive loads. Furthermore, while examining the impact of low-side device gate resistance on crosstalk, the influence of low-side switch parasitic emitter inductance becomes evident, and the optimal values for these parameters are determined. The model is generic and applicable to any IGBT or power MOSFET, because crosstalk happens during turn-ON process which the turn-ON model of both devices is similar, with the specific case study in this article being the IXGH60N60C2 and IRFP460.

Index Terms—Crosstalk, double-pulse test, insulated gate bipolar transistor (IGBT), modeling, parasitic elements.

I. INTRODUCTION

HIGH voltage/power insulated gate bipolar transistors (IGBTs) find extensive use across various applications in power converters. They are commonly utilized in traction systems [1], high-voltage dc/dc converters [2], pulsed power supplies [3], HVdc circuit breakers [4], [5], [6], and solid-state transformers [7], [8]. The reliability of power converters holds paramount significance, given their crucial role in various industrial applications. Previous research reveals that approximately 34% of power converter failures can be attributed to issues related to IGBTs [9], [10]. Although power IGBTs demonstrate robustness under severe conditions such as short circuit

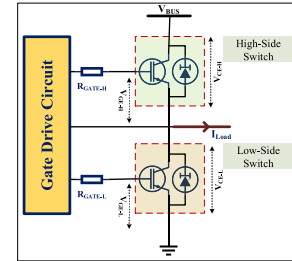


Fig. 1. Generic schematic of phase-leg.

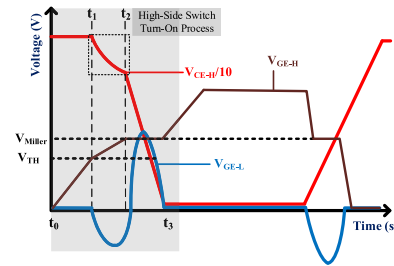


Fig. 2. Gate-emitter voltage of the lower side device during an upper device transient.

faults [11], [12], they face multiple reliability challenges during operation. Consequently, improving the reliability of IGBTs and gaining a comprehensive understanding of the causes of failures in both normal and harsh conditions have become focal points of recent research efforts.

When IGBTs are employed within a phase-leg configuration as it is shown in Fig. 1, the occurrence of shoot-through currents resulting from dc-link short-circuits can prove detrimental to the IGBTs. The phase-leg configuration is prevalent in various power converter designs, including single/multiphase inverters and dc/dc resonant converters. Even when intentional dead-time is incorporated between IGBT operations within the phase-leg, the risk of shoot-through currents remains significant. These shoot-through currents are primarily attributed to crosstalk or parasitic turn-ON phenomena, wherein the activation of the upper device inadvertently triggers the lower side device, as illustrated in Fig. 2. As a consequence, both devices conduct simultaneously for a brief period, allowing the shoot-through current to pass within the phase-leg. Crosstalk is induced by the voltage variation in the Miller capacitance of the low-side device during the transition of the high-side device from the

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off-state to the on-state and vice versa. When the high-side device shifts from the off to the on state, a positive voltage surge occurs in the gate-emitter voltage of the low-side device, as depicted in Fig. 2. When the peak value of the gate-emitter voltage (V_{GE-LP}) surpasses the device's gate threshold voltage (V_{TH}), it enters the active region, permitting saturated current to flow through the IGBTs within the phase-leg structure. The additional power loss resulting from crosstalk can elevate the device's junction temperature. In certain scenarios, the short-circuit current magnitude can be several times that of the device's nominal current, posing a potential risk of device failure.

Various solutions to mitigate crosstalk and parasitic turn-ON have been documented in prior research. One proposed approach in [13] involves applying a negative bias to the gate-emitter terminals of the devices during the off-state, which helps reduce the gate-emitter peak voltage in crosstalk. However, the cost associated with adding an isolated dc power supply to the system may not be feasible in many applications, and it is incompatible with boot-strap gate drive systems. Furthermore, negative bias alone may not be sufficient to prevent parasitic turn-ON in numerous cases. Gate impedance regulation has been explored in [14], [15], and [16], aiming to optimize gate impedance to minimize the impact of Miller capacitance on the gate side during crosstalk. While these solutions are practical, their effectiveness is limited to specific conditions. In addition, various innovative gate drive systems have been proposed in [17], [18], and [19], with a primary focus on the gate side. However, the extensive investigation into the impact of gate resistance alone and emitter parasitic inductance on crosstalk has not been thoroughly examined. Crucially, the behavior of crosstalk is closely tied to the gate resistance value and emitter's inductance. Therefore, before developing a crosstalk suppression strategy, the adoption of a predictive approach for determining V_{GE-LP} in crosstalk appears imperative.

Various models for crosstalk are available in the literature, ranging from simpler analytical models [15], which are easy to characterize and overlook parasitic elements, to more complex ones [20], which incorporate power parasitic components. However, the latter model's limitation lies in its neglect of the transient speed of the high-side device during the turning-ON process. The most comprehensive model in the literature is presented in [21]. This model focuses on the power path circuit, and the load type considered is resistive, which causes the following shortcomings.

- 1) A primary limitation lies in the fact that resistive loads are not universally applicable, given that real-world loads mainly are inductive or resistive-inductive characteristics.
- 2) The second constraint is that in the case of a purely resistive load, the adverse impact on crosstalk is overlooked. The negative spike occurs exclusively during inductive load switching, resulting from residual energy in the parasitic inductances, potentially leading to gate-oxide failure (GOF). This implies that the model is applicable only in very specific scenarios. However, when dealing with an inductive load, the negative fluctuation becomes noticeable and can be taken into account.

This article seeks to present a thorough model incorporating parasitic elements. The analysis will focus on evaluating the model concerning parasitic parameters, followed by the derivation of the crosstalk transfer function. In the proposed model, the inductive load switching of the device is considered, which highlights the impact of the gate side parasitic elements in crosstalk. Subsequently, the investigation will delve into the influence of gate resistance and the interdependence of the low-side switch emitter parasitic inductance on crosstalk. The discussion will then address pertinent considerations for designers, culminating in the introduction of an optimal design in order to reduce the crosstalk risk (CTR). To propose an optimal design, meticulous consideration should be given not only to the CTR but also to the reverse gate-emitter voltage. This reverse voltage has the potential to cause GOF and subsequent device malfunction. Careful attention to these factors is crucial to ensuring the reliability and proper functioning of the device in question.

This article is structured as follows. Section II delves into the analysis of the test circuit and presents a comprehensive model for crosstalk. In Section III, we explore the impact of gate resistance on CTR and investigate the interplay between low-side switch emitter inductance and gate resistance. Section III proposes designing considerations for the phase-leg structure, accompanied by equations for PCB trace inductance and resistance. Section IV puts forth an optimal design for the phase-leg configuration, outlining recommended regions for gate resistance and emitter inductance. Section V provides insight into an experimental test setup designed to validate the simulation and analytical model findings. Finally, Section VI succinctly concludes this article.

II. TEST CIRCUIT AND PROPOSED CROSSTALK MODEL

This article introduces a novel and comprehensive crosstalk model, which distinguishes itself from existing literature models for IGBTs in several key aspects. These distinctions are outlined as follows.

- 1) First, by including all parasitic elements, both internal and external, the model is generalized for all types of IGBTs.
- 2) Second, a double-pulse test with an inductive load is employed to introduce a novel and accurate model for predicting crosstalk.
- 3) Third, the impact of the partly parabolic curve of V_{CE-H} (approximated linearly in this study) is analyzed, and the parasitic element that plays a critical role in determining its value is investigated.

A. Double-Pulse Test Setup

To examine the interplay between parasitic parameters and the crosstalk phenomenon, Fig. 3 illustrates a double-pulse test circuit incorporating a range of parasitic parameters. This test circuit simulates the inductive load switching behavior of the devices [24]. This test involves the application of two closely spaced voltage pulses to the device, simulating turn-ON and turn-OFF events. The first pulse initiates the turn-ON process,

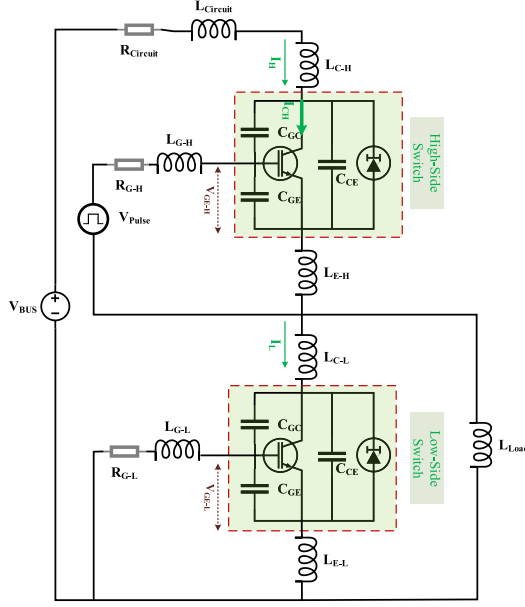


Fig. 3. Double-pulse testing setup incorporating diverse parasitic elements.

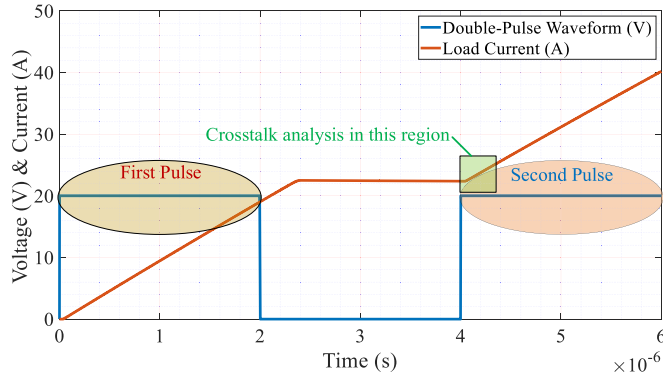


Fig. 4. Double pulse test waveform and load current.

subjecting the device to a fast rise in voltage. This pulse simulates the practical scenario of the device transitioning from an off-state to an on-state. The second pulse, closely following the first, emulates the second turn-ON process with inducing voltage again, representing the transition from an off-state to an on-state, again. Hence, when dealing with an inductive load, its behavior can be likened to that of a current source. Fig. 4 demonstrates the double pulse waveform used in the simulation and the experimental tests.

B. Mathematical Model

In this model, the parasitic elements are represented as $L_{Circuit}$, L_{C-H} , L_{E-H} , L_{G-H} , L_{C-L} , L_{E-L} , L_{G-L} , C_{GC} (referred to as the Miller capacitance), C_{GE} , C_{CE} , $R_{Circuit}$, R_{G-H} , and R_{G-L} . These resistances collectively account for parasitic, internal, and external gate-side resistances. Parasitic inductances of the emitter terminal encompass both the internal inductance of the IGBTs (L_{int}) and the inductance associated with the PCB traces (L_{trace}). The summation of these components contributes to the

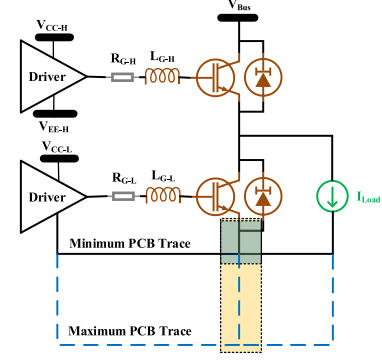


Fig. 5. Concept of L_{E-L} in phase-leg structure.

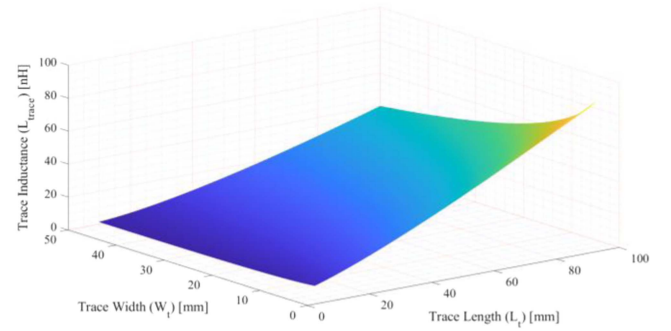


Fig. 6. Inductance value based on length and width.

overall parasitic inductance within the circuit L_{E-L} . Altering L_{E-L} is achievable by adjusting the length and width of the trace. Fig. 5 illustrates the circuit, indicating how L_{E-L} is determined, whereas Fig. 6 showcases the inductance value by modifying both the trace length and width. The L_{E-L} can be varied from 4 to 85 nH depending on the length and width of the PCB trace. The length directly influences the trace inductance, while the width has an inverse impact. Each individual element may exert varying impacts on crosstalk, necessitating an in-depth analysis. Therefore, this article specifically investigates the influence of low-side gate resistance and emitter inductance.

1) *Stage 1* [t_1 – t_2 in Fig. 2]: At time t_1 , the gate-emitter voltage of the high-side device (V_{GE-H}) rises to the threshold voltage (V_{TH}), initiating the turn-ON process of the high-side device. The channel current (I_{CH}) of the high-side switch begins to increase from zero. Concurrently, the current amplitude flowing through the low-side IGBT diode decreases, indicating a positive increment of I_L . Consequently, voltage induced on L_{E-L} emerges. V_{CE-L} remains constant, preventing the generation of Miller current, and the transfer current of C_{GC} does not propagate into C_{GE} . Thus, crosstalk is exclusively influenced by the induced voltage source $L_{E-L} dI_L/dt$. Throughout this stage, V_{GE-L} undergoes negative fluctuations, marking the appearance of the initial negative spike [25]. In addition, this negative spike is unavoidable due to the emitter inductance of the IGBT package and PCB trace.

The corresponding circuit equation and the equivalent circuit of the low-side IGBT driver during this stage are illustrated in

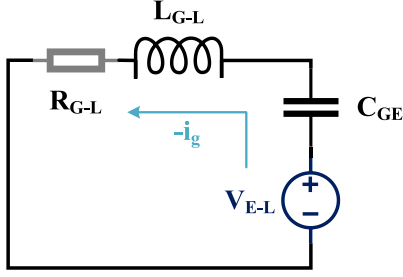


Fig. 7. Equivalent circuit of low-side device during stage 1.

Fig. 7

$$V_{E-L} + V_{GE-L1} + L_{G-L} \frac{di_g}{dt} + R_{G-L} i_g = 0 \quad (1)$$

where

$$\begin{cases} i_g = C_{GE} \frac{dV_{GE-L1}}{dt} \\ V_{E-L} = L_{E-L} \frac{dI_L}{dt} \end{cases} \quad (2)$$

Combining with (1) and (2), V_{GE-L} can be written as

$$V_{GE-L1} = -V_{E-L} + \frac{V_{E-L}}{a_2 - a_1} (a_2 e^{-a_1 \omega_1 t} - a_1 e^{-a_2 \omega_1 t}) \quad (3)$$

where

$$\begin{cases} a_1 = \varepsilon_1 - \sqrt{\varepsilon_1^2 - 1} \\ a_2 = \varepsilon_1 + \sqrt{\varepsilon_1^2 - 1} \\ \omega_1 = \frac{1}{\sqrt{L_{G-L} C_{GE}}} \\ \varepsilon_1 = \frac{R_{G-L}}{2} \sqrt{\frac{C_{GE}}{L_{G-L}}} \end{cases} \quad (4)$$

In the above-mentioned expression, ω_1 represents the natural frequency, and ε_1 signifies the damping coefficient. The damping coefficient ε_1 escalates proportionally with the augmentation of R_{G-L} and C_{GE} . Consequently, the rate of change of V_{GE-L} in (3) decelerates. During this phase, expressions for I_L and I_{CH} can be formulated as follows:

$$I_L = I_{CH} - I_{Load} \quad (5)$$

$$I_{CH} = gfs \times (V_{GE} - V_{TH}) \quad (6)$$

where gfs is the transconductance coefficient. By combining (2), (5), and (6), V_{E-L} can be expressed as

$$\begin{aligned} V_{E-L} &= L_{E-L} \times gfs \times \frac{dV_{GE-H1}}{dt}; \\ L_{E-L} &= L_{int} + L_{trace}. \end{aligned} \quad (7)$$

In accordance with (7), V_{E-L} increases with the rise of L_{E-L} or dV_{GE-H}/dt , causing V_{GE-L} in (1) to become more negative. As the current in low-side IGBT body diode reaches zero, the reverse recovery process initiates. This phase concludes when I_L nears the peak value of the reverse recovery current for the body diode.

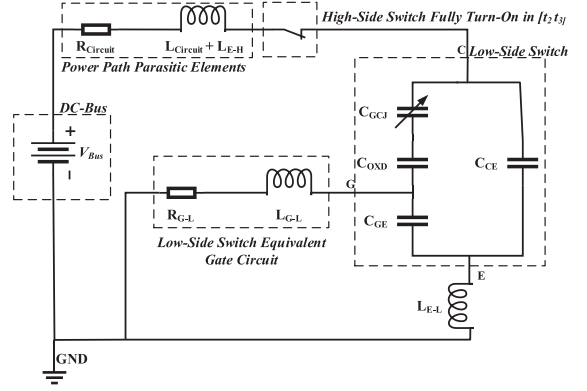


Fig. 8. Model of the phase-leg configuration during stage 2.

2) Stage 2 [t_2 - t_3 in Fig. 2]: From t_2 to t_3 , the gate-collector capacitor of the low-side IGBT is charged. Therefore, considering the C_{GC} as a nonlinear capacitor of the IGBT is mandatory. Equation (8) is taken from IGBT physical model [26], [27] and describes the behavior of the depletion gate-drain region. Miller capacitance of IGBT is created by the variable capacitance C_{GDI} , representing the depletion zone, and a constant capacitance C_{OXD} , representing the oxide zone in series mode as shown in Fig. 8

$$C_{GCJ} = A_{GC} \sqrt{\frac{q \times N_B \times \varepsilon_{si}}{2(V_{CE} - V_{GE})}} \quad (8)$$

Hence, C_{GC} can be found as

$$C_{GC} = \begin{cases} \frac{C_{OXD} \times C_{GCJ}}{C_{OXD} + C_{GCJ}} & \text{if } V_{CE} > V_{GE} \\ C_{OXD} & \text{if } V_{CE} \leq V_{GE} \end{cases} \quad (9)$$

In the expression (8), gate-drain area A_{GD} and dielectric constant ε_{si} , can notably alter the modulation of C_{GCJ} within the depletion region. However, in this research V_{CE} has directly impact on the value of C_{GCJ} , and as a result C_{GC} .

At time point t_2 , the reverse voltage across the body diode of the low-side device grows. This upward trend in voltage results in an increase in V_{CE-L} , the voltage across the low-side IGBT collector-emitter terminals. Consequently, there is a rise in Miller current flowing out of C_{GC} , the gate-controlled capacitor. This elevated Miller current intensifies the rate of increase in V_{GE-L} , the voltage across the low-side IGBT gate-emitter terminals, leading to a positive fluctuation in V_{GE-L} . This intricate interplay highlights the interdependence of these voltage parameters within the low-side IGBT, showcasing how changes in one aspect can have cascading effects, influencing the overall behavior of the device. By modeling the low-side IGBT as three capacitors such as Fig. 8, in stage 2, Kirchhoff's equations can be written and the gate-emitter voltage likewise in [21] can be written as

$$V_{GE-L2} = \frac{C_{GC} V_{Bus} e^{-s \times (t_3 - t_2)} (e^{s \times (t_3 - t_2)} - 1) \times (R_G + L_G s)}{s(t_3 - t_2) \times (D_4 s^4 + D_3 s^3 + D_2 s^2 + D_1 s + 1)} \quad (10)$$

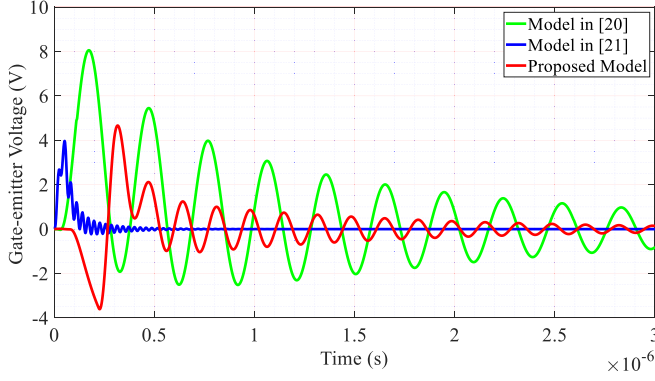


Fig. 9. Comparison between the latest model in the literature and the new model in this article.

The coefficients of the (10) from [21] are

$$\begin{aligned}
 D_4 &= (L_{Circuit} + L_{E-H}) ((L_{EL} + L_{G-L}) (C_{CE}C_{GE} \\
 &\quad + C_{GC}C_{GE}) + C_{CE}C_{GC} \times L_{G-L}) \\
 &\quad + C_{GC}C_{GE} \times L_{E-L}L_{G-L} \\
 D_3 &= C_{CE}C_{GC} (L_{Circuit} + L_{E-H}) R_{G-L} \\
 &\quad + C_{CE}C_{GC}L_{G-L}R_{Circuit} \\
 &\quad + C_{CE}C_{GE} (L_{Circuit} + L_{E-H}) R_{G-L} \\
 &\quad + C_{CE}C_{GE}L_{G-L}R_{Circuit} \\
 &\quad + C_{GC}C_{GE} (L_{Circuit} + L_{E-H}) R_{G-L} \\
 &\quad + C_{GC}C_{GE}L_{G-L}R_{Circuit} \\
 &\quad + C_{CE}C_{GE}L_{E-L}R_{Circuit} \\
 &\quad + C_{GC}C_{GE}L_{E-L}R_{Circuit} \\
 &\quad + C_{GC}C_{GE}L_{E-L}R_{G-L} \\
 D_2 &= C_{CE}C_{GC}R_{Circuit} R_{G-L}C_{CE}C_{GE}R_{Circuit} R_{G-L} \\
 &\quad + C_{GC}C_{GE}R_{Circuit} R_{G-L} \\
 &\quad + C_{CE}L_{Circuit} + C_{GC}L_{Circuit} \\
 &\quad + C_{GC}L_{G-L} + C_{GE}L_{G-L} \\
 &\quad + C_{GE}L_{E-L} \\
 D_1 &= R_{Circuit} (C_{CE} + C_{GC}) + R_{G-L} (C_{GC} + C_{GE}) \quad (11)
 \end{aligned}$$

Fig. 9 demonstrates the difference between the latest model in the literature [21] and the novel model introduced in this article. Experimental results in Section V will show that the new model works accurately in terms of predicting CTR and GOF. The most substantial difference between the proposed model and existing ones is that most of them ignore the impact of crosstalk on GOF. GOF is very probable in inductive load switching of the device and has been scrutinized by the model proposed in this article. It is noteworthy to mention that the crosstalk takes place in turning-ON process of the devices. Based on the investigations of extant studies such as [22] and [23] the turning-ON procedure of IGBTs and MOSFETs are the same. Accordingly, the proposed crosstalk model is also applicable for power MOSFETs.

TABLE I
MODEL VALUES AND PARAMETERS

$R_{Circuit}$	1 Ω
$L_{Circuit} + L_{C-H}$	80 nH
Device Part Number IGBT	IXGH60N60C2
Device Part Number MOSFET	IRFP460
$L_{E-H} (L_{int} + L_{trace})$	3 nH
L_{C-L}	3 nH
$L_{G-H} \& L_{G-L}$	3 nH
I_{Load}	20 A
V_{Bus}	400 V
R_{G-L}	[5 – 25 Ω]
$L_{E-L} (L_{int} + L_{trace})$	[5 – 50 nH]

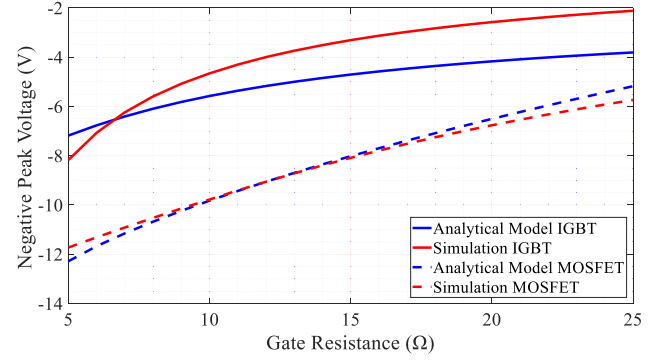


Fig. 10. Impact of gate resistance on negative spike with definite emitter inductance of 5 nH for IGBT and 17 nH for MOSFET with Table I values using PSPICE.

III. CODEPENDENCY OF GATE RESISTANCE AND EMITTER INDUCTANCE

As demonstrated in (4), the direct impact of crosstalk is influenced by the low-side gate resistance through the damping coefficient. Adjusting R_{G-L} will directly impact the negative spike of the crosstalk. When R_{G-L} reaches a sufficiently large value, the rate of change in V_{GE-L} approaches zero, indicating that V_{GE-L} is maintained at its initial voltage. Nevertheless, elevating R_{G-L} , will result in a decrease in the switching speed. Consequently, it is not advisable to significantly increase R_{G-L} as a preventive measure against the negative spike, as it may lead to potential system instability.

To assess the accuracy of the proposed model, simulations are conducted in PSPICE.

Table I demonstrates the model values based on valid structures and datasheets. The influence of crosstalk becomes apparent with alterations in R_{G-L} .

Increasing the R_{G-L} value, while keeping the value of L_{E-L} unchanged, leads to a noticeable reduction in the negative spike, as indicated by (3) and (4). The impact of R_{G-L} on the peak of the negative spike is illustrated in Fig. 10. The error between analytical model and simulation is due to the linearity approximation of parameters and also the average value of C_{GC} . However, this error is not dominant and is negligible regarding the achieved results. As a result, as the value of R_{G-L} increases, the V_{GE-LP} also rises, increasing the likelihood of crosstalk occurrences as demonstrated in Fig. 11.

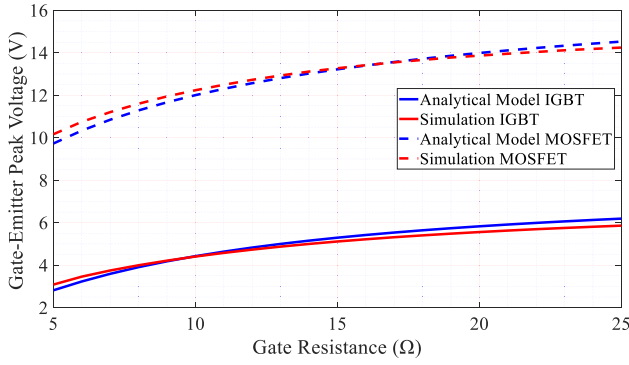


Fig. 11. Impact of gate resistance on crosstalk with definite emitter inductance of 5 nH for IGBT and 17 nH for MOSFET with Table I values using PSPICE.

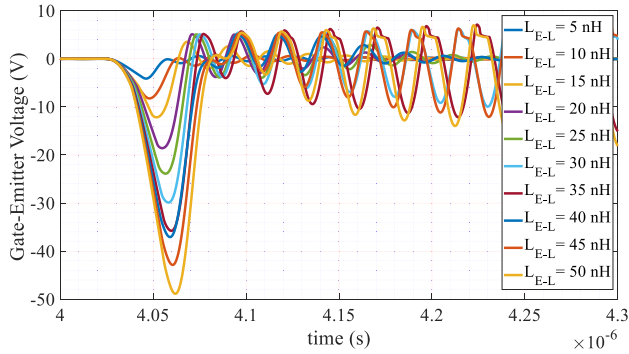


Fig. 12. Impact of varying low-side emitter inductance on negative spike for the IGBT case study.

A. Codependency of Low-Side Parasitic Inductance

In addition to the gate resistance, there exists a codependency that highlights the significant impact of the low-side switch emitter inductance on the system. Equation (7) explicitly reveals that the voltage source of the low-side IGBT emitter is directly influenced by L_{E-L} . As per (3), it is evident that a decrease in gate resistance necessitates a decrease in low-side emitter inductance. This implies that the PCB track should have a short length and sufficient width to effectively handle the passing current. In a specific application where the speed of the switches holds significantly important, maintaining small L_{E-L} is crucial. Ineffective design with high parasitic inductance poses a heightened risk of malfunction and instability in the phase-leg. On the contrary, increasing the value of L_{E-L} exacerbates the negative spike, intensifying the severity of crosstalk. Consequently, concerns about crosstalk and potential parasitic turn-ON diminish as the negative spike becomes more pronounced. However, this negative fluctuation can lead to problems in IGBT, which in Section III-B the consequences is discussed.

As an illustration, opting for a gate resistance of 4 Ω leads to a variation in emitter inductance from 5 to 50 nH, with other specifications from Table I showcasing the corresponding outcomes. Fig. 12 illustrates this observation. Clearly, as L_{E-L} increases, crosstalk fades away. However, beyond a certain threshold, the system becomes unstable. As depicted in

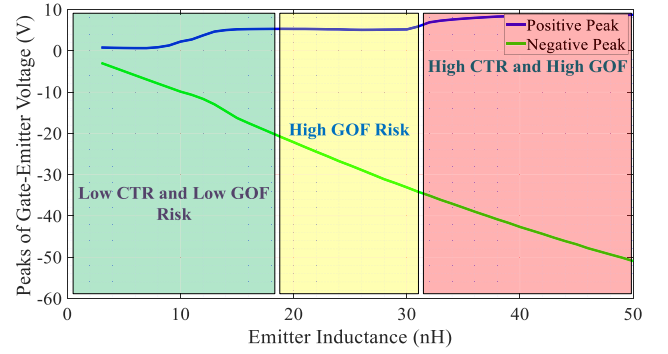


Fig. 13. Impact of varying low-side emitter inductance on crosstalk for the IGBT case study.

Fig. 13, beyond 30 nH, the peak voltage of V_{GE-LP} intensifies, indicating that the designer should steer clear of this region and minimize the inductance of the printed circuit board (PCB).

B. Negative Consequence of Reverse Gate-Emitter Voltage

As discussed in Section III, particularly in Section III-A, augmenting the low-side emitter inductance results in an increase in the negative peak of the gate-emitter voltage as shown in Fig. 12. While this phenomenon can be advantageous for the CTR, it is essential to note that an excessively negative peak may have detrimental consequences, potentially leading to the destruction of the IGBT or power MOSFET. If the reverse gate-emitter voltage exceeds 25 V, there is a risk of GOF, where the resonance between L_{E-L} and C_{GE} can compromise the isolation of the gate layer, culminating in irreversible damage [28]. Careful monitoring and control of these voltage parameters are imperative to prevent such deleterious effects on the device.

C. Designing Considerations

In this section, an examination is conducted on sets of considerations for the design of the phase-leg structure. The primary focus lies on the evaluation of parasitic inductance, primarily influenced by the PCB trace and the internal inductance of discrete IGBT or module, with specific details provided in the datasheet.

To establish a reasonable range for the inductance of the PCB power trace, it is imperative to calculate the trace width. The IPC-2221 standard is employed for determining trace width, resulting in a specific inductance based on the provided values. Equation (12) provides the necessary PCB trace area (A) [29], where I_{trace} represents the current flowing through the trace, T_r signifies the rise in temperature of the PCB, and constants (k , b , and c) are defined by the IPC-2221 standard. Following the identification of the trace area, the trace width in relation to the thickness of the trace (T) is offered

$$A = \frac{I_{trace}}{(k(T_r^b))^{\frac{1}{c}}}$$

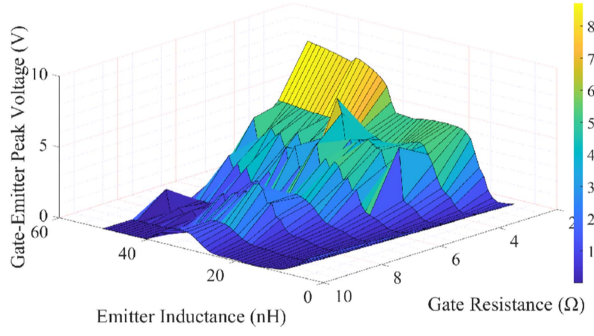


Fig. 14. Impact of emitter inductance and gate resistance on crosstalk for the case study presented in Table I.

$$W_t = \frac{A}{T \times 1.378}. \quad (12)$$

In accordance with the findings presented in [30], (13) furnishes the PCB trace inductance (L_{trace}) as a function of substrate height (H), trace width (W_t), trace thickness (T), trace length (L_t), vacuum permeability (μ_0), and relative permeability (μ_r).

$$L_{\text{trace}} \cong \frac{\mu \times \left(H + \frac{T}{2}\right) \times L_t}{W_t}, \quad \mu = \mu_0 \mu_r. \quad (13)$$

Moreover, the resistance of the gate-side circuit is pivotal and necessitates careful consideration. The resistance of the PCB trace can be ascertained through the following procedure. In (14), the ρ is the resistivity of the trace, which for copper generally is $1.7e-6 \Omega/\text{cm}$ and α is the temperature coefficient, which for copper is $3.9e-31/^\circ\text{C}$ [31]

$$R_{\text{trace}} = \rho \times \frac{L_t}{T \times W_t} \times (1 + \alpha \times (\theta - 25)). \quad (14)$$

Alongside the trace resistance, internal and external resistances exist in the gate driver IC and the designer's resistor, contributing to the adjustment of the switches' speed. Fig. 11 illustrates that as the gate resistance value increases, the likelihood of crosstalk also intensifies.

To sum up this section, based on the aforementioned findings and applying crosstalk model equations, achieving an optimal design is imperative to minimize crosstalk and enhance the system's reliability. The subsequent section will delve into the exploration of an optimal design for the phase-leg configuration.

IV. OPTIMAL DESIGN

As elucidated in the preceding sections, various factors contribute to crosstalk. Gate resistance and low-side emitter inductance exert significant influence on crosstalk, delineating specific regions that demand caution from the designer. It is crucial to note the interdependence between gate resistance and parasitic emitter inductance.

Fig. 14 illustrates a 3-D surface, showcasing the direct impact of these elements on crosstalk and the interdependency existing between them for values between 10 and 50 nH as an example. Fig. 14 demonstrates that crosstalk is more likely to occur when the gate resistance is very low with a large emitter inductance.

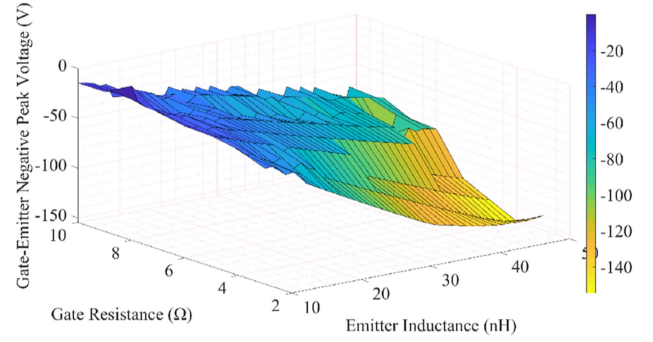


Fig. 15. Impact of emitter inductance and gate resistance on negative peak of gate-emitter voltage for the case study presented in Table I.

Conversely, as the gate resistance increases, crosstalk diminishes. However, it is essential to maintain a sufficient amount of emitter inductance to ensure that short circuit failure does not occur in the circuit. Unlike Fig. 11, this 3-D surface represents cases with low gate resistance, likely under 7Ω , combined with high emitter inductance. This figure clearly illustrates the relationship between low gate resistance and high emitter inductance. Therefore, when high emitter inductance is needed to diminish crosstalk, then the value of the gate resistance should exceed 10Ω in order to ensure crosstalk does not happen in the circuit. After certain value of gate resistance which in this case study is 7Ω , with rising the gate resistance with a high value of emitter inductance, CTR increases likewise Fig. 11, and the peak of gate-emitter voltage starts to increase.

Fig. 15 demonstrates the impact of high emitter inductance and low gate resistance on the negative peak gate-emitter voltage. In contrast to Fig. 14, the negative peak voltage trend is similar to that in Fig. 13. With low values for the gate resistance, the probability of GOF increases, even with lower values of emitter inductance.

Typically, the internal inductance values of IGBTs range from 3 to 13 nH and the parasitic values can be extracted from the device datasheet [32]. When considering the addition of PCB trace inductance to the internal inductance, this combined value can vary between 4 and 85 nH. Consequently, selecting an appropriate gate resistance is imperative, and it should ideally exceed 7Ω to mitigate the risk of potential failures.

As a demonstration, the R_{G-L} sets to 10Ω is used in many applications and reasonable value. For this specific gate resistance, there should be an optimum inductance. According to Fig. 16, increasing the emitter inductance initially results in a reduction of crosstalk; however, beyond a certain point, the reverse peak voltage can pose a risk to the switch, particularly in this case, an IGBT. Therefore, the optimal range, as indicated by Fig. 17, falls between 3 and 50 nH. This range provides a favorable balance that designers can readily achieve. Our model demonstrates an effective predictability of crosstalk. Prior to any design implementation, designers can proactively consider various elements to mitigate crosstalk and minimize the risk of potential short-circuit failures.

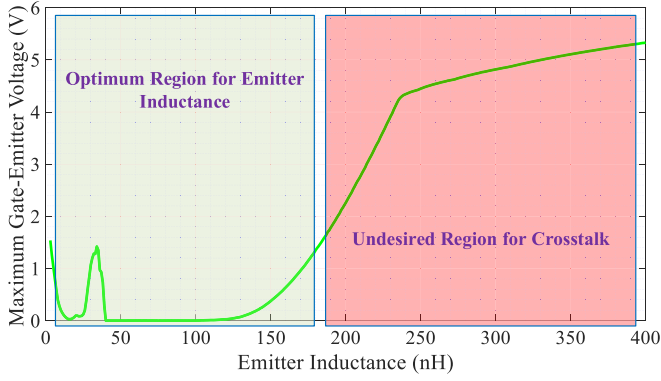


Fig. 16. Analysis of gate-emitter voltage on crosstalk where R_{G-L} is set to 10Ω .

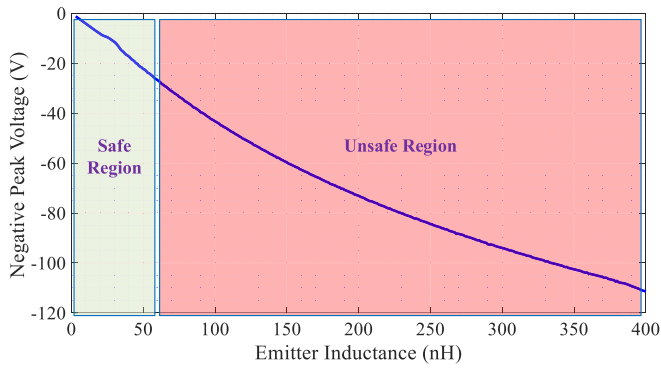


Fig. 17. Analysis of gate-emitter voltage on negative spike where R_{G-L} is set to 10Ω .

V. EXPERIMENTAL RESULTS

To validate the research outcomes via simulations, a phase-leg structure is instantiated, and experiments are carried out. Figs. 18 and 19 present an overview and block diagrams of the test setup, respectively.

In the conducted double-pulse test, the experimental case mirrors the simulated conditions, maintaining consistent specifications. The test procedure unfolds as follows.

- 1) *Initialization*: The trigger unit initiates a low-level signal, activating the low-side device driver. Consequently, the low-side device assumes the off-state, primed for the upcoming test.
- 2) *High-Side Device Activation*: A high-level signal with a pulse width of $10 \mu s$ is generated by the trigger signal, activating the high-side device. Notably, both switches remain in the off-state prior to the commencement of this high-level signal.

This double-pulse test configuration allows for the meticulous examination of the switching characteristics and performance of the phase-leg structure under controlled conditions.

Table II presents the values used in the experimental setup for this case study, along with measurements for the parasitic elements. The values of L_{G-H} and L_{G-L} extracted from datasheet, which is the internal parasitic inductances, and the L_{E-L} and $L_{Circuit}$ were calculated from the length and width of the PCB and the ESL.

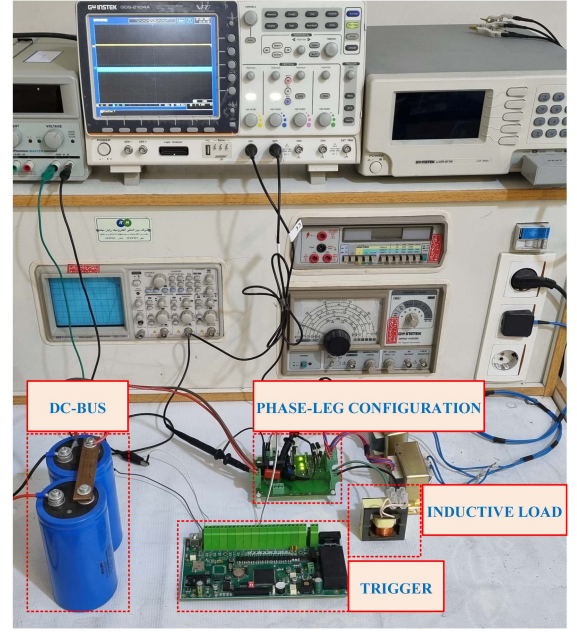


Fig. 18. Test setup.

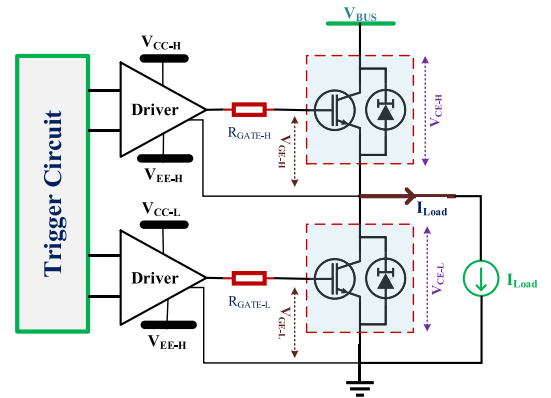


Fig. 19. Block diagram of the test circuit.

TABLE II
EXPERIMENTAL PARAMETERS VALUE

$R_{Circuit}$	1Ω
$L_{Circuit} + L_{C-H}$	80 nH
L_{C-L}	5 nH
$L_{E-H} (L_{int} + L_{trace})$	5 nH
$L_{G-H} \& L_{G-L}$	3 nH
Device Part Number	IXGH60N60C2
I_{Load}	20 A
V_{Bus}	400 V
$L_{E-L} (L_{int})$	$[10 - 33 \text{ nH}]$
$R_{G-H} \& R_{G-L}$	20Ω

Fig. 20 illustrates the initial waveform capturing the turn-ON process of the high-side switch and the gate-emitter voltage of the low-side device. Notably, a discernible negative fluctuation is observed during the initial time interval, followed by subsequent positive fluctuations as predicted by the analytical equations mentioned earlier. In this particular case study, the gate resistance is set at 20Ω , and the emitter inductance is established at 10 nH . In this particular case study, the threshold voltage

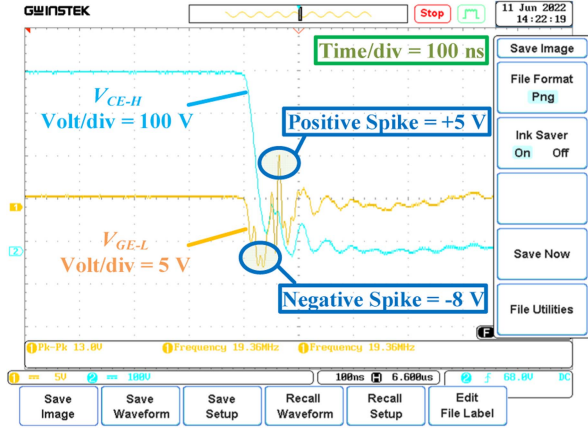


Fig. 20. Low-side switch waveforms in the high-side switch turning-ON process.

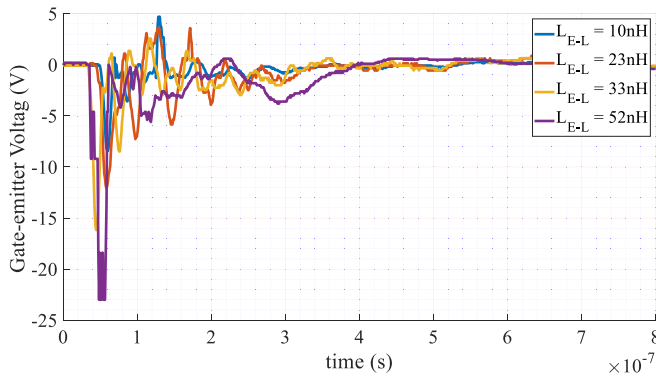


Fig. 21. Comparison among three emitter inductances.

(V_{TH}) of the IGBT is typically 5 V, as indicated in the datasheet. The presence of this positive spike poses a potential danger, especially if the low-side IGBT turns ON, leading to a short-circuit failure. Consequently, there is a crucial need to augment the low-side emitter inductance to prevent the occurrence of a possible short circuit in the phase-leg. As illustrated in Fig. 20.

To validate the simulations and analytical models, experimental tests were conducted. Two excess inductances were introduced to the initial value of the low-side emitter inductance to assess the impact of L_{E-L} on CTR. The positive value of the gate-emitter voltage (V_{GE-L}) exhibited a decrease as L_{E-L} increased. Conversely, the absolute value of the reverse gate-emitter voltage experienced a sharp rise, posing a potential risk of GOF in the phase-leg. Therefore, prudent consideration is essential during the phase-leg design to mitigate both CTR and GOF risks. Fig. 21 presents the waveforms obtained from the experimental tests, confirming the validity of the analytical model and simulations.

To verify the model for the impact of gate resistance, tests have been conducted to demonstrate that increasing gate resistance with a low value of emitter inductance can increase crosstalk (CTR). Fig. 22 shows the variation of gate-emitter voltage versus gate resistance with a 10 nH emitter inductance. As the gate resistance increases, the switching speed of the device decreases and the negative voltage reduces. However, the peak gate-emitter voltage rises, thereby increasing the likelihood of CTR. Fig. 23

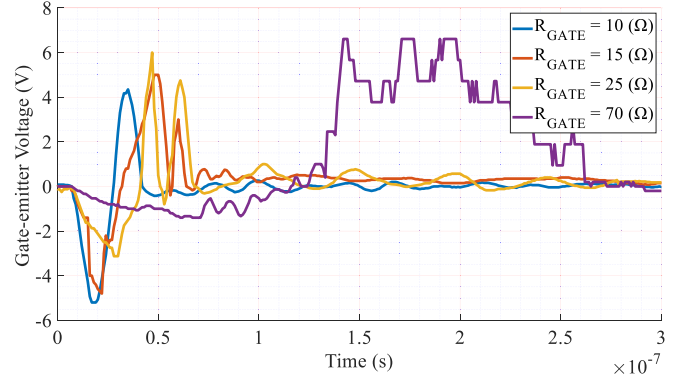


Fig. 22. Varying the gate resistance with an absolute L_{E-L} of 10 nH.

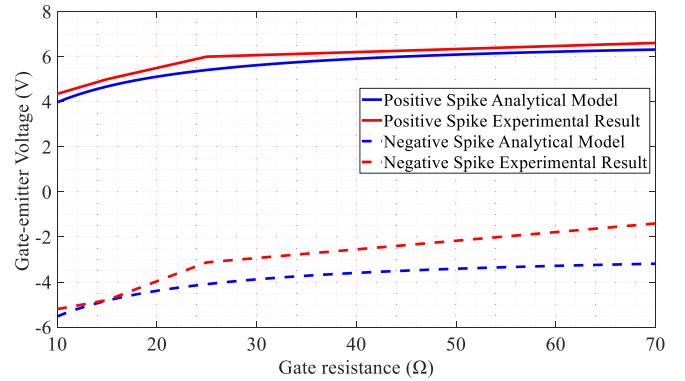


Fig. 23. Verification of the model based on experimental results for IGBT case study.

shows the minimum and maximum voltages obtained from the experimental setup and compared with the analytical model. When compared to the introduced model, the results confirm that the model is accurate and applicable to other part numbers for both IGBTs and MOSFETs.

VI. CONCLUSION

This article is dedicated to a comprehensive exploration of crosstalk phenomena within the context of IGBTs configured in a phase-leg structure with inductive load characteristics. The results of this study are summarized in the following.

- 1) Introduced a comprehensive model highlighting the role of parasitic elements in the phase-leg setup, and developed a mathematical circuit model for detailed analysis with a double-pulse test methodology for examining device switching characteristics.
- 2) Emphasized the importance of maintaining high enough gate resistance to prevent instability, besides the crucial role of substantial low-side emitter inductance in minimizing CTR and GOF.
- 3) This article suggested an optimal range for both gate resistance and low-side emitter inductance as a co-dependence element in the circuit with an optimal design case with experimental test results.
- 4) This article recommended the negative bias for the gate side of IGBTs, with considering other elements in order to avoid possible GOF.

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